

Machine learning for complex-junction guidance

A scenario-based navigation optimisation for high-cost, long-tail driving situations

Lead Product Manager | initiated and led the project

Anonymised case study | six-month delivery

2–3%

of junctions are complex

48%

higher off-route rate

28%

fewer motorway
detours



The decision arrives at speed

At a motorway split, one missed exit can add tens of kilometres before the next safe correction.

2–3%

of junctions
are complex

+48%

off-route rate vs.
ordinary junctions

MATURE PRODUCT

Long-tail junctions became the next product frontier

Core utility

The product completes the main navigation journey



Scenario quality

Long-tail situations receive guidance designed for their specific risk

I proposed this project because complaints persisted and another rule would not solve the long tail.

Rules could not scale to new junction types

THE OLD SYSTEM

rule 1	road class
rule 2	lane count
rule 3	connection angle
rule 4	city exception
rule 5	new junction patch

One rule: about 3 person-days

Earlier rules could silently exclude the same junction, so a new patch might never fire.

WHY THIS SLICE FIT MACHINE LEARNING

01 Clear label

People could judge whether a junction was genuinely difficult.

02 Human ground truth

A consistent annotation policy turned judgement into training data.

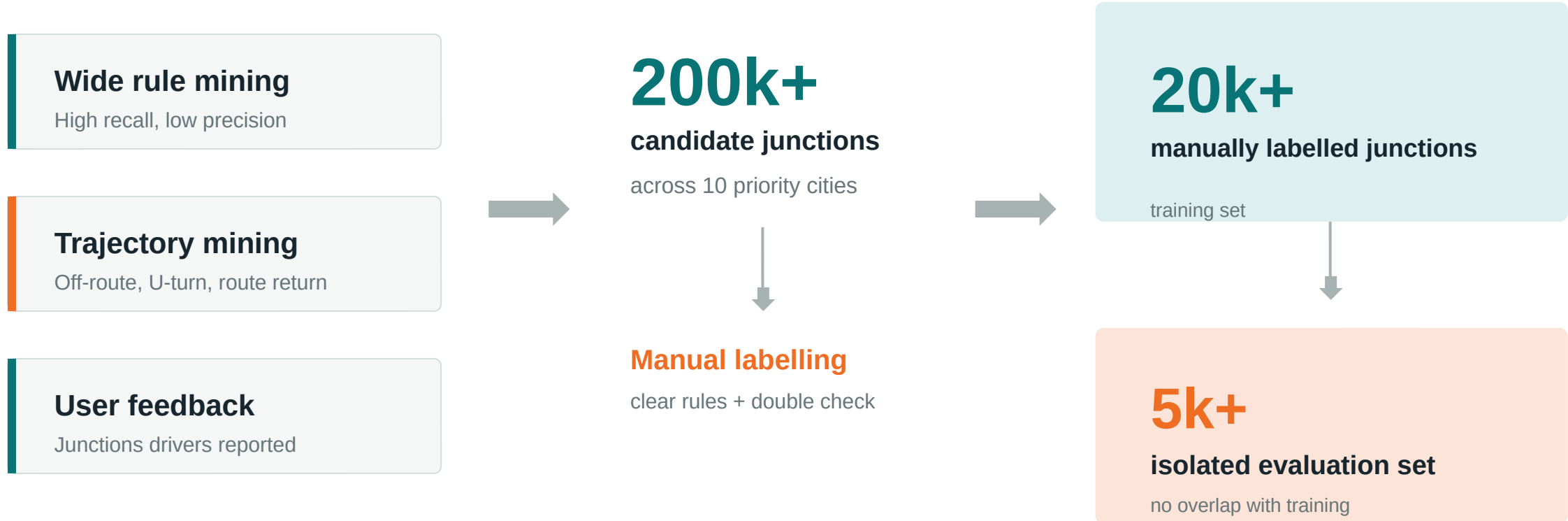
03 Signals in the data

Map geometry and real driver trajectories captured different evidence.

A bounded, measurable scenario made ML useful without pretending to learn navigation quality end to end.

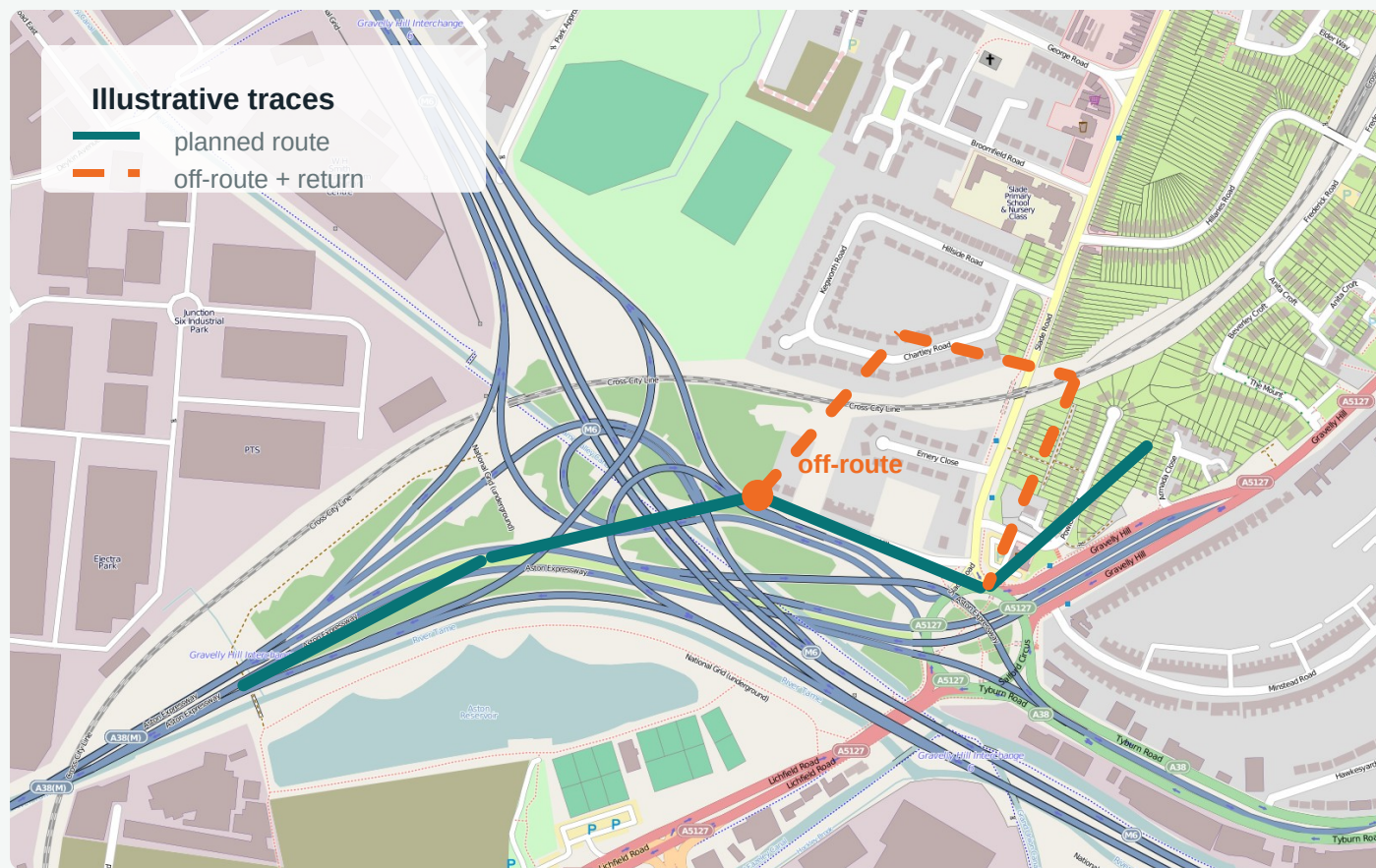
Three candidate sources built a broader labelled set

Candidate sources



The three sources created candidates. Human annotation created ground truth.

Physical-world context improved model quality



01 Trajectory behaviour as evidence

I proposed weighting junctions when drivers went off-route, made a U-turn, or returned to the planned route.

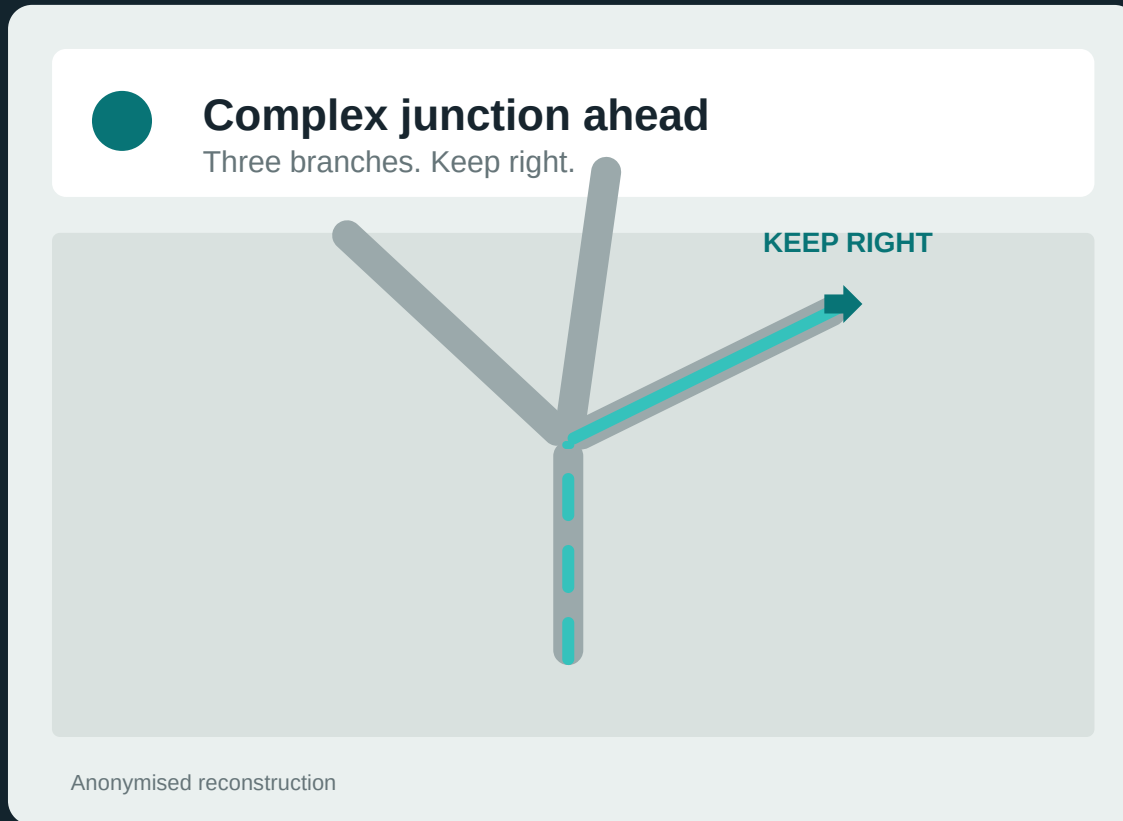
02 City and road-class splits

The breakdown exposed elevated-road cases where positioning drift looked like junction complexity. We adjusted the mining weight instead of rewarding the wrong signal.

The data said 'complex junction'.
The physical world said 'signal delay'.

Confidence became product policy

The model estimated confidence. I decided how the product should respond.

**HIGH**

85%+ precision

Voice + junction diagram**MEDIUM**

specific

Voice names the junction type**UNCERTAIN**

non-directive

Soft warning preserves recall**NEGATIVE**

no prompt

The product stays silent

A lower-confidence detection received a lower-cost prompt, not the same instruction with less certainty.

The model reduced errors where guidance was live

Relative change in off-route rate

MODEL
DISCOVERED

-18%

-discovered junctions

KNOWN
DIFFICULT

-48%

Known difficult junctions

28% fewer motorway detours

Measured on controlled-access motorways

ROLLOUT EVIDENCE

2 months

A/B test over one release

15+ cities

from 1% traffic to full release

Millions

of navigation users covered

The global off-route rate did not move because complex junctions were only 2–3% of all junctions.

MY OWNERSHIP

Project origin | labelling policy | feature hypotheses | confidence thresholds | rollout

The confidence-tiered product policy later transferred to metro and indoor positioning.